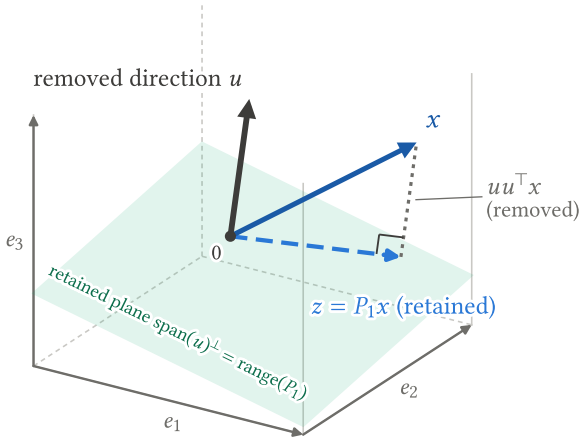
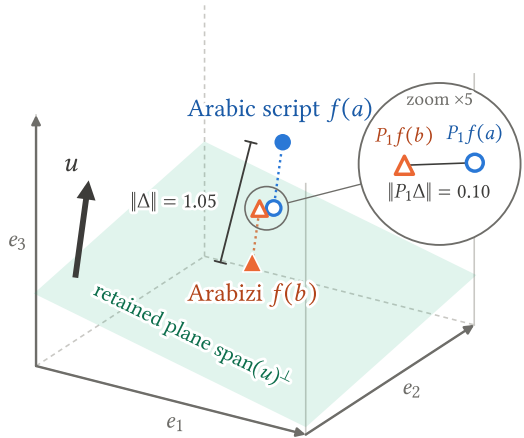
(a) Orthogonal decomposition $x = P_1 x + uu^\top x$



(b) Arabic-script and Arabizi pair, before and after P_1



original vector
retained (projected)
removed direction u
removed component / projection guide
Arabic script
Arabizi
after P_1 (open)

Schematic: constructed vectors in $\mathbb{R}^3$, not measured embeddings